

Initial assessment of BDS-3 precise point positioning service on GEO B2b signal

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Abstract

The China BeiDou global navigation satellite system (BDS-3) has started to provide free and open precise positioning services to users in China and surrounding areas since Jul 2020 on the GEO B2b signal. The initial assessment of BDS-3 PPP-B2b products and the PPP performance over a period of near five months from Jul 26, 2020 to Dec 19, 2020 was carried out in this work. Taking the GBM products as a reference, the average signal-in-space ranging error (SISRE) of the disseminated orbit and clock products is approximately 82.3 cm (RMS) and 3.9 cm (STD) for BDS-3 satellites, and about 135.8 cm (RMS) and 7.5 cm (STD) for GPS. The average positioning error RMS at six permanent stations in China is about 2.1 (1.8) and 2.6 (2.1) cm in the horizontal and vertical components respectively for BDS-only (BDS + GPS) static PPP. In the kinematic PPP mode, the average positioning error (95%) at the same six stations is about 21.5 (15.2) and 33.4 (30.3) cm in the horizontal and vertical component respectively for BDS-only (BDS + GPS), with an average convergence time of 17.4 (16.2) min. In addition, two vehicle-based kinematic PPP tests show average position error (95%) of about 23.5 (18.6) and 48.8 (37.1) cm in horizontal and vertical respectively, and an average convergence time of 10.4 (14.2) min for BDS-only (BDS + GPS) PPP. The assessment results are overall in accordance with the official claim of centimeter-level in static mode and decimeter-level in kinematic mode from BDS-3 PPP service.

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1. Introduction

The BeiDou global navigation satellite system (BDS-3), with nominal space constellation consisting of 3 geostationary earth orbit (GEO) satellites, 3 Inclined Geo-Synchronous Orbit (IGSO) and 24 Medium Earth Orbit (MEO) satellites, was announced the official launch on Jul 31, 2020. Besides the basic positioning, navigation and timing functions (i.e., the RNSS functions), the BDS-3 additionally provides global short message communica-

tion (GSMC) and international search and rescue (SAR) services to global users, as well as the regional short message communication (RSMC), the satellite based augmentation system (SBAS), and precise point positioning (PPP) services in China and surrounding areas (Yang et al., 2020).

Different from the Real Time Kinematic (RTK) technique, PPP can achieve decimeter-level, centimeter-level or even millimeter-level positioning accuracy by the processing of carrier phase measurements of a stand-alone receiver (Zumberge et al., 1997), but precise satellite orbit and clock products are indispensable. The International GNSS Service (IGS) analysis centers (ACs) have been routinely estimating and releasing the precise products in post-processing mode based on a global GNSS network since

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1990s (Johnston et al., 2017), which, however, cannot meet the requirements in real-time applications, motivating the launch of the IGS real-time service (IGS-RTS) pilot project in 2007 (IGS, 2011). IGS-RTS ACs generate the real-time precise satellite products and disseminate them following Networked Transport of RTCM via Internet Protocol (NTRIP). The quality of IGS-RTS products has been continuously improved afterwards. The standard deviation (STD) of the signal-in-space ranging error (SISRE) are reported to be from 1.3 to 5.5 cm for products from different IGS-RTS ACs (Zhang et al., 2018).

Users are required to have network accesses in order to receive the NTRIP product data stream (e.g., IGS-RTS) to realize the real-time PPP (RT-PPP), which is however not realistic in some scenarios, for example, on the ocean. For these uses, the satellite can be used as a transmitter to disseminate satellite products, namely, the satellite-based augmentation PPP. There have been many commercial agencies nowadays providing the global satellite-based augmentation PPP services via commercial satellites such as the Inmarsat. In recent years, in order to foster innovation in both consolidated and emerging markets, several GNSS systems have been working to embed the PPP service in the navigation systems. For example, the Japanese QZSS is already providing centimeter-level augmentation service (CLAS) over Japan (Cabinet Office, 2019). The European Galileo promises to offer a free global high-accuracy (HA) service on E6-B signal to enable PPP with error of less than two decimeters (Borio et al. 2020). The Galileo E6-B interface control document (ICD) is already defined and under final consolidation. The Russian GLO-NASS System Differential Correction Monitoring (SDCM) also has a plan to provide PPP services initially on a regional basis but to expand to global afterwards (Stupak, 2013).

The BDS-3 PPP-B2b ICD was released in July 2020, which claims to have the ability to provide kinematic decimeter-level and static centimeter-level PPP to users in China and surrounding areas. The BDS-3 PPP service signal, PPP-B2b, transmitted by the BDS-3 GEO satellites, serves as the data broadcasting channel for correction parameters, such as satellite precise orbit and clock offset parameters, of BDS-3 and other GNSSs (only GPS in current stage) (China Satellite Navigation Office, 2020). Lu et al. (2020) decoded the PPP corrections based on a software-defined receiver (SDR) and concluded that the PPP-B2b signal can stably provide PPP services in the Asia-Pacific region. There have been a bunch of studies discussing the performance of BDS-3 RNSS. For example, the SISRE root mean square (RMS) estimated from the broadcast ephemeris was reported to be about 0.5 m, with continuity and availability of about 99.99% and 99.78% (Guo et al., 2019). The timing accuracy is better than 19.1 ns (95%). The precision of the post-processing orbit can reach centimeter-level with the inclusion of inter-satellite links (ISL), and the average satellite clock offset uncertainty of MEO satellites is 1.55 ns (Yang et al., 2020). Improvements

in both accuracy and convergence time are found by combining BDS-3 with BDS-2 in PPP (Shi et al., 2020; Shi et al., 2021; Cao et al., 2021), but the inter-system bias between BDS-2 and BDS-3 needs careful consideration (Jiao et al., 2019). However, there are few publications on the evaluations of BDS-3 PPP-B2b products and the corresponding PPP performance. Nie et al. (2021) conducted an initial assessment of BDS-3 PPP-B2b products and PPP performance at four permanent stations, but the assessment only covered a 3-day period.

This work is therefore carried out to comprehensively evaluate the BDS-3 PPP-B2b products (referred as B2b products hereafter) from the aspect of the availability and accuracy. The performance of PPP based on B2b products in both static and kinematic mode are then assessed. The paper is organized as follows: the B2b products will be evaluated in Section 2. The static and kinematic PPP performance will be discussed in Section 3 and Section 4, respectively, followed by the conclusions in Section 5.

2. PPP-B2b product evaluations

BDS-3 PPP service information is broadcasted on the open service signal PPP-B2b, with a center frequency of 1207.14 MHz and a bandwidth of 20.46 MHz. The first three GEO satellites, located at 80°E, 110.5°E and 140°E respectively, only transmit PPP-B2b products on the I-component with a symbol rate of 1000 symbols per second. Each message has a length of 486 bits, wherein the highest 6 bits indicate the message type as defined in Table 1, the lowest 24 bits are CRC, and the remaining 456 bits are the message data.

The Femtomes FB680 receiver, which is placed at the roof of a building near Wuhan University in Wuhan, China, as shown in Fig. 1, was used to receive the BDS-3 PPP-B2b signal from Jul 26, 2020 (day of year, DOY, 208) to Dec 19, 2020 (DOY354). The satellite orbit corrections, clock corrections, and DCB were then decoded and recovered to satellite orbit and clock files with the help of the received navigation message (B1C-CNAV1 for BDS-3 and LNAV for GPS).

Details about the coding, receiving, and using of the PPP-B2b products can be referred to the ICD. Here we will only give a brief description of the algorithm for using B2b products.

For the application of the DCB correction, the formula can be written as,

$$l_{sig} = l_{sig} - DCB_{sig} \quad (1)$$

where l_{sig} and l_{sig} denote the observed code value of sig signal before and after applying the DCB corresponding to the signal with value of DCB_{sig} .

The satellite position $\vec{X}_{orbit}$ can be recovered following,

$$\vec{X}_{orbit} = \vec{X}_{broadcast} - \delta\vec{X} \quad (2)$$

Table 1
BDS-3 PPP-B2b defined message type.

Message Types	Information Content
1	Satellite mask
2	Satellite orbit correction and user range accuracy index
3	Differential code bias (DCB)
4	Satellite clock correction
5	User range accuracy index
6	Clock correction and orbit correction – combination 1
7	Clock correction and orbit correction – combination 2
8–62	Reserved
63	Null message

where $\vec{X}_{broadcast}$ denotes the satellite position calculated by the broadcast ephemeris where the IOD should match with the IODN of the orbit correction message. $\delta\vec{X}$ represents the satellite position correction, and can be calculated by,

$$\vec{e}_{radial} = \frac{\vec{r}}{|\vec{r}|} \quad (3)$$

$$\vec{e}_{cross} = \frac{\vec{r} \times \dot{\vec{r}}}{|\vec{r} \times \dot{\vec{r}}|} \quad (4)$$

$$\vec{e}_{along} = \vec{e}_{cross} \times \vec{e}_{radial} \quad (5)$$

$$\delta\vec{X} = [\vec{e}_{radial} \ \vec{e}_{along} \ \vec{e}_{cross}] \delta\vec{O} \quad (6)$$

where $\vec{r}$ and $\dot{\vec{r}}$ denote the satellite position and velocity vector of broadcast ephemeris. $\vec{e}_{radial}$, $\vec{e}_{along}$, and $\vec{e}_{cross}$ are direction unit vector in the radial, along and cross component respectively. $\delta\vec{O}$ is PPP-B2b orbit correction including information in order of radial, along and cross component.

The satellite clock offset $t_{satellite}$ can be recovered as,

$$t_{satellite} = t_{broadcast} - \frac{C_0}{c} \quad (7)$$

where $t_{broadcast}$ is the satellite clock offset calculated from the broadcast ephemeris. c_0 denotes the clock correction parameter obtained from PPP-B2b message, and c is the velocity of light.

2.1. Product availability

During our test period, only messages with type 1 to type 4 were received. The nominal update cycle of the orbit correction, clock correction and DCB for each satellite is 48, 6, and 48 s, respectively. It takes 48 s to receive a completed record for all satellites. The messages cover BDS-3 satellites with Pseudo-Random Noise (PRN) code from C19 to C46, and GPS satellites with PRN from G01 to G32. The availability of each kind of product is calculated on a daily basis as the proportion between the received epoch number and the theoretical epoch number (duration divided by the nominal update cycle). Days when the receiver experienced an outage or technical issues (about 42 days) which causes the absence of products are excluded in the statistic.

The averaged availability of B2b products over the test period for each satellite is illustrated in Fig. 2. For BDS-3, C31 is a test IGSO satellite with no correction information broadcasted. The average orbit corrections, clock corrections and DCB availability of all other satellites are about 36.13, 35.21 and 36.30%, respectively. The relatively low rates are due to the fact that the B2b products are generated based on a regional network with stations mainly in China. Compared to MEO satellites, IGSO satellites (C38–C40) have longer periods tracked by the regional network, resulting in higher availability of 70–80% for all products. For GPS satellites, during our test period, only orbit and clock corrections are received, with an average availability of 30.00 and 26.60%, respectively. Corrections for G23 (newly launched Block III-A satellite) cannot be received until Oct. 22 (DOY296), resulting in a very low availability of 9.80% for orbits, and 8.44% for clocks.

We also calculate the percentage of epochs with more than five available BDS-3 or five GPS satellites in the B2b products which are generally believed to be necessary for reliable PPP solutions. The availability is approximately 98.39% (orbit), 97.46% (clock) and 99.74% (DCB) for BDS-3 satellites, while is about 98.00% (orbit) and 96.48% (clock) for GPS satellites.



Fig. 1. The receiver (left) and antenna (right) used for BDS-3 PPP-B2b signal reception.

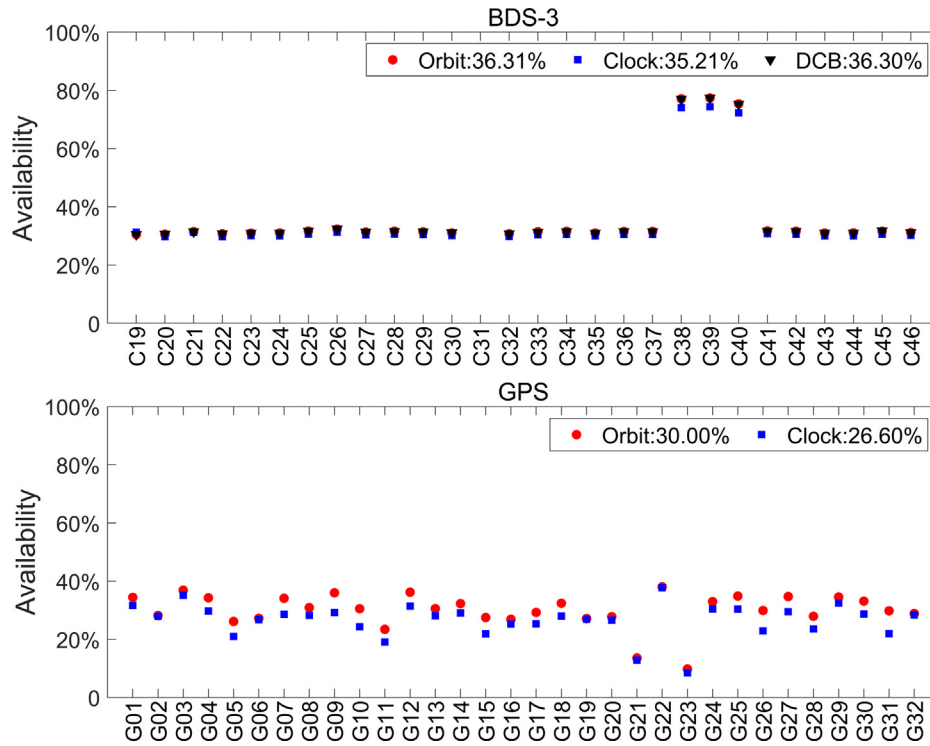


Fig. 2. The availability of PPP-B2b decoded satellite orbit corrections, clock corrections and DCB over the test period.

2.2. Product accuracy

Taking the rapid multi-GNSS products (GBM) released by GFZ (GeoForschungsZentrum) as reference (Deng et al. 2017), the recovered B2b orbit and clock products during our test period are evaluated in this section. GBM products are interpolated based on the Lagrange method to the B2b orbit and clock epochs. Orbit differences in ECEF (Earth-Centered Earth-Fixed) coordinate system are converted to the radial (R), along (A) and cross (C) component, and RMS of the differences are then calculated. For clock products, differences for each satellite are calculated and the average differences of all satellites at each epoch (the clock datum) are then removed. Both RMS and standard deviation (STD) of clock differences are calculated where RMS and STD are quality indicators for pseudorange and carrier-phase measurement processing, respectively. In addition to the orbit and clock accuracy statistics, the SISRE that combines the impact of orbit and clock errors on the users is also estimated as (Montenbruck et al., 2015),

where R , A , and C denote the orbit difference in radial, along and cross components in meter, respectively. Clk represents the clock difference in meter.

The daily orbit, clock and SISRE evaluation results are illustrated from Figs. 3–5, and statistic results are summarized in Table 2. The gaps in Figs. 3–5 are caused by the receiver outage or technical issues as mentioned in Section 2.1. As for comparisons, in Table 2, we also present the evaluation results of products from two IGS-RT ACs, the Wuhan university (WUC) (Zhang et al., 2016; Shi et al., 2019) and Centre National d’Etudes Spatiales (CNC) (Laurichesse et al., 2009), that produce both GPS and BDS-3 real-time orbit and clock products during the same evaluation period.

The average B2b BDS-3 orbit error RMS is about 10.2, 31.8 and 27.6 cm in the radial, along and cross component, compared to 9.4, 32.1 and 9.4 cm for WUC, and 4.1, 12.2 and 5.4 cm for CNC products. Orbit errors can be partly assimilated by the clock estimates, resulting in approximate 2.936 ns (RMS) and 0.243 ns (STD) for B2b clock products, compared to 1.615 and 0.263 ns for WUC, and

$$SISRE = \begin{cases} \sqrt{(0.98 \cdot R - Clk)^2 + \frac{1}{49}(A^2 + C^2)} & GPS \\ \sqrt{(0.99 \cdot R - Clk)^2 + \frac{1}{126}(A^2 + C^2)} & BDS(GEO/IGSO) \\ \sqrt{(0.98 \cdot R - Clk)^2 + \frac{1}{34}(A^2 + C^2)} & BDS(MEO) \end{cases} \quad (8)$$

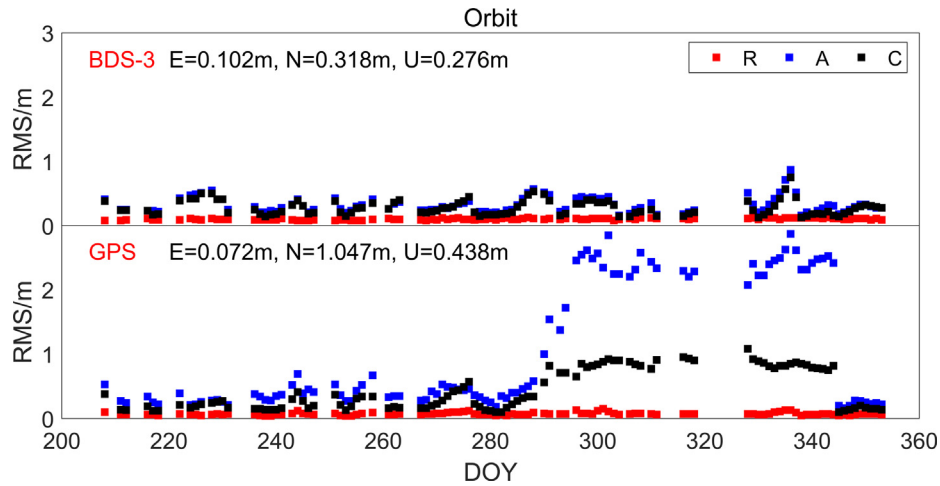


Fig. 3. B2b orbit product error RMS.

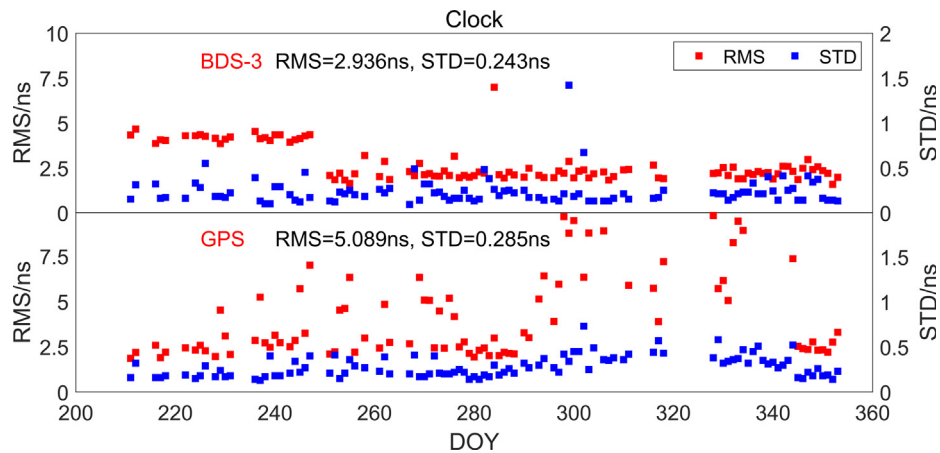


Fig. 4. B2b clock product error RMS and STD.

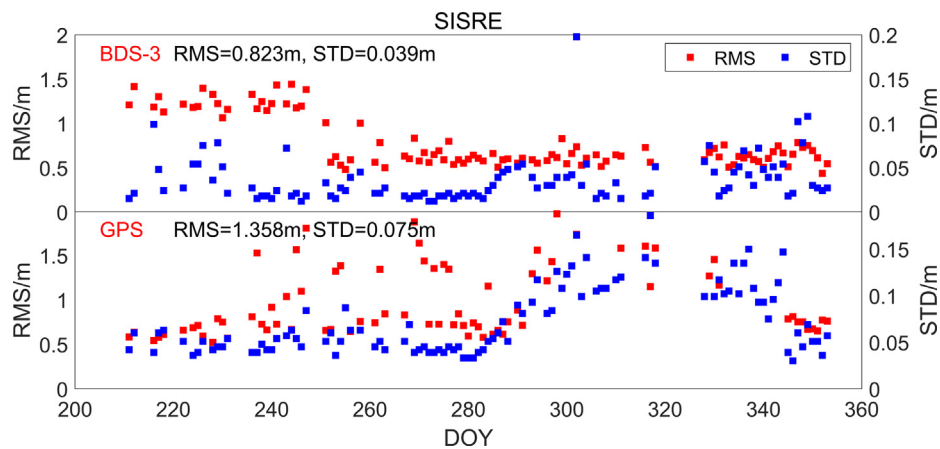


Fig. 5. B2b product SISRE RMS and STD.

1.168 and 0.189 ns for CNC. The SISRE STD of B2b is, however, only 3.9 cm, smaller than both WUC (5.1 cm) and CNC (4.2 cm), but the SISRE RMS are obviously larger, reaching 82.3 cm. The smaller SISRE STD is mainly

resulted from the use of ISL observations in the B2b product determination.

For GPS satellite, B2b orbit quality experiences an obvious degradation in the along component after Oct 16, 2020

Table 2
Statistics of B2b, WUC and CNC product accuracy.

		Orbit (m)			Clock (ns)		SISRE (m)	
		Radial	Along	Cross	RMS	STD	RMS	STD
GPS	B2b	0.072	1.047	0.438	5.089	0.285	1.358	0.075
	WUC	0.017	0.055	0.027	0.498	0.113	0.114	0.025
	CNC	0.026	0.036	0.025	0.494	0.105	0.111	0.016
BDS-3	B2b	0.102	0.318	0.276	2.936	0.243	0.823	0.039
	WUC	0.094	0.321	0.094	1.615	0.263	0.443	0.051
	CNC	0.041	0.122	0.054	1.168	0.189	0.345	0.042

(DOY290), and this abnormality lasts for nearly two months when GPS orbits returned to normal on Dec 11, 2020 (DOY346). The average B2b orbit error for GPS is therefore significantly larger than WUC and CNC, especially in the A and C components, with RMS of 7.2, 104.7 and 43.8 cm in the radial, along, and cross components, respectively. The B2b GPS clock errors and SISRE are also affected by the orbit abnormality, with RMS of 4.089 ns and 135.8 cm, and with STD of 0.285 ns and 7.5 cm for clock and SISRE, respectively.

3. Static PPP performance

Observations at six stations in China were processed with the decoded B2b products in simulated real-time mode, which means the observations and satellite products are read from files instead of real-time data streams, and the data processing is in one-way forward. The six stations contain one IGS MGEX (Multi-GNSS Experiment) station (WUH2) (Montenbruck et al., 2017), and five iGMAS (international GNSS Monitoring and Assessment System) (Jiao et al., 2011) stations (BJF1, KUN1, LHA1, SHA1 and XIA1) as shown in Fig. 6. For BDS-3, the observations at B1I and B3I frequencies are used with the B2b DCB product applied to align with the B2b BDS-3 clock product which are generated based on the ionosphere-free combina-

tion of B1C and B2a observations. The data processing strategy is summarized in Table 3.

The ionosphere-free combination of code and phase measurements at time t , between satellite s and receiver r , namely, $P_r^s(t)$ and $L_r^s(t)$ in meters, used in PPP in this study can be expressed as,

$$\begin{cases} P_r^s(t) = \rho_r^s(t) + c \cdot t_r(t) - c \cdot t^s(t) + ISB_r + m_r^s(t) \cdot ztd_r(t) + \varepsilon_p(t) \\ L_r^s(t) = \rho_r^s(t) + c \cdot t_r(t) - c \cdot t^s(t) + ISB_r + m_r^s(t) \cdot ztd_r(t) + \\ \lambda_1 \cdot N_r^s(t) + \lambda_1 \cdot \phi_r^s(t) + \varepsilon_L(t) \end{cases} \quad (9)$$

where ρ denotes the geometric distance between the satellite and receiver with the phase center offset (PCO), phase center variations (PCV), relativistic effect, and tide effect corrected. t_r and t^s are receiver and satellite clock offset where t^s is fixed to the PPP-B2b-recovered parameter. ISB_r represents the inter-system-bias. For BDS-only PPP, ISB_r is set to be zero. For GPS + BDS PPP, ISB_r for GPS is set to be zero while ISB_r for BDS is estimated as a daily constant. m_r^s is the mapping function and ztd_r is the zenith tropospheric delay. λ_1 denotes the wavelength at the first frequency for GPS or BDS satellites. N_r^s and ϕ_r^s represent the ambiguity and phase wind-up effect, respectively. ε_p and ε_L are items including noise and unmodeled errors for code and phase respectively.

GNSS observations were processed in a daily manner. The position estimates at the last epoch in each day are compared with the ‘true’ positions which are estimated in post-processing mode. The post-processing mode means that the GBM products rather than the B2b products were fixed and the iterative least square method rather than the one-way forward filter method was applied for the parameter estimation. However, it still needs to be noted that the positions estimated in post-processing mode may be biased in the same manner as real-time solutions. The average static PPP position errors in the horizontal and vertical components for all test stations are presented in Fig. 7 and are summarized in Table 4. We can find that the horizontal and vertical position errors are smaller than 5 and 10 cm respectively for almost all solutions in both BDS-only and BDS + GPS static PPP, which is in accordance with the claim of centimeter-level positioning accuracy for static PPP in the PPP-B2b ICD. Gaps in the result time series, for example, near DOY 320, are caused by the absence of B2b products which is due to the outages of the receiver

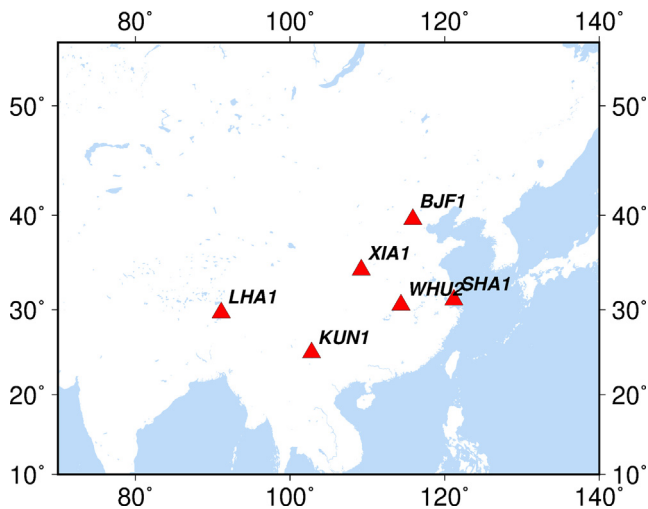


Fig. 6. Geographical distribution of PPP stations.

Table 3
GNSS Data processing strategy.

Observations	Observations Sampling interval Elevation cutoff Weighting strategy	Ionosphere-free combination of L1/L2 for GPS and B1I/B3I for BDS-3 30 s 7° Elevation-dependent: 1 for elevation (E) >30°, otherwise 2sin(E) a-priori sigma: 3 mm and 0.3 m for the phase and code observation, respectively Weight ratio is 1:1 for GPS and BDS-3 satellite
Models	Satellite PCO/PCV Receiver PCO/PCV Phase wind-up effect Tide model Tropospheric model Mapping Function	igs14.atx (Schmid et al., 2016) igs14.atx for GPS observations use GPS values for BDS-3 observations Corrected Solid tide; Polar tide; Ocean tide; (IERS Conventions, 2010) ZHD corrected by GPT (Boehm et al., 2007) and Saastamoinen (Saastamoinen, 1972) model GMF (Boehm et al., 2006)
Parameter Estimation	Satellite orbit and clocks Receiver position ZWD Receiver clock Ambiguity	Fix to PPP-B2b products Static mode: daily constants Kinematic mode: epoch-wise parameter without constraints between epochs Random walk with constrains of 1.5 mm/ $\sqrt{h}$ White noise Float

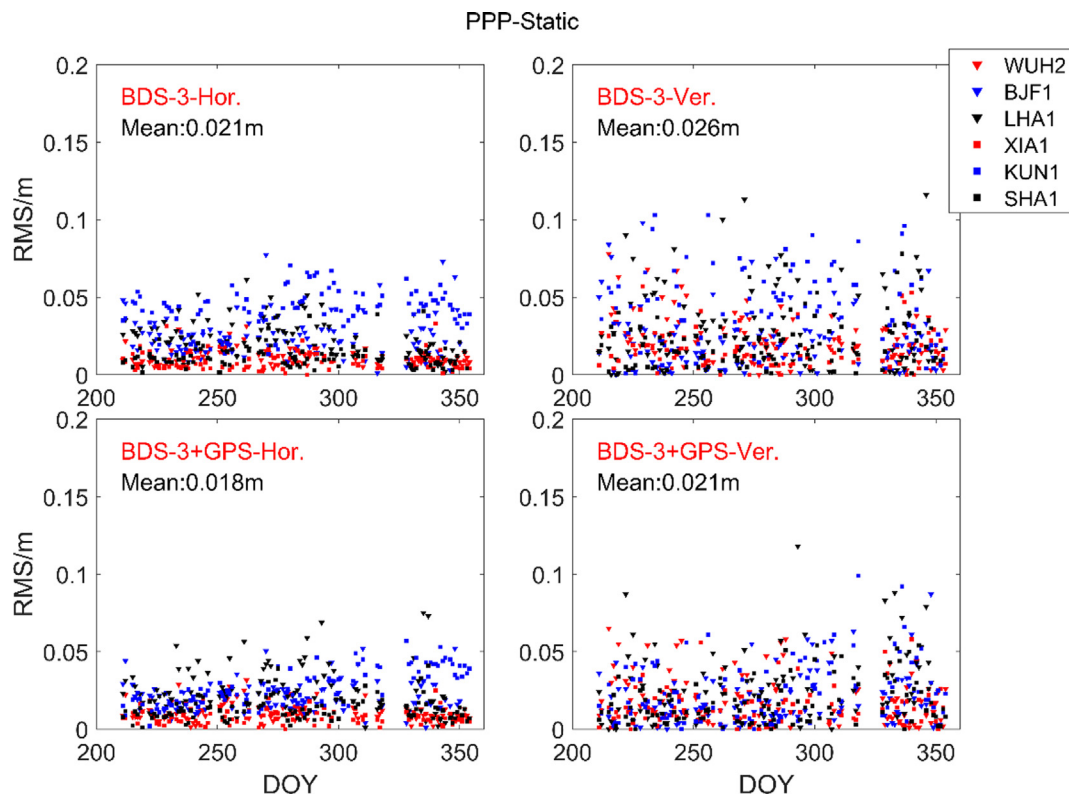


Fig. 7. Static PPP position error for BDS-only (top) and BDS + GPS (bottom) in the horizontal (left) and vertical (right) component.

used for collecting B2b products. The combination of BDS and GPS can significantly improve the positioning accuracy. The average position error RMSs for all stations by combining BDS-3 and GPS (BDS + GPS) are approximately 1.8 and 2.1 cm in the horizontal and vertical component respectively, compared to 2.1 and 2.6 cm by using BDS-only observations, with improvement of 16.7% and 23.8%.

4. Kinematic PPP performance

4.1. Simulated kinematic PPP at permanent stations

Observations at the same six permanent GNSS stations were also processed in simulated kinematic PPP mode, which means that positions are estimated as epoch-wise parameters without constraints between epochs. Similar

Table 4
Static PPP position error at permanent stations.

Stations	BDS-only		GPS + BDS	
	Hor. (m)	Ver. (m)	Hor. (m)	Ver. (m)
BJF1	0.011	0.024	0.011	0.022
LHA1	0.025	0.023	0.022	0.020
KUN1	0.025	0.031	0.027	0.027
WHU2	0.009	0.016	0.008	0.014
SHA1	0.042	0.040	0.026	0.024
XIA1	0.013	0.020	0.014	0.018
Mean	0.021	0.026	0.018	0.021

to Section 3, the ‘true’ positions are also from the static mode by using GBM satellite products. The PPP convergence time is defined following the system designed objectives as the cost time when the position errors firstly reach and keep smaller than 0.3 (0.2) and 0.6 (0.4) m in the horizontal and vertical component, respectively, for BDS-only (BDS + GPS) for at least 10 consecutive minutes. By excluding the period before the convergence, the 95% smallest ranked epoch-wise position error is taken as the accuracy indicator.

The daily position error (95%) time series in the horizontal and vertical components for BDS-only and GPS + BDS at the six stations are shown in Fig. 8 and statistic results are summarized in Table 5. The position accuracy is overall improved during the test period for BDS-only PPP, especially in the vertical component. The average position error of all stations is about 21.5 and 33.4 cm in the horizontal

and vertical components, with 82.3% and 96.9% smaller than 0.3 and 0.6 m. By combining GPS and BDS-3 observations, the average kinematic PPP error decreases to 15.2 and 30.4 cm. The abnormality in the B2b GPS products after DOY290 has obvious impacts on the BDS + GPS PPP solutions which can be visually seen from Fig. 8. The percentage of position errors smaller than 0.2 and 0.4 m is therefore about 81.0% and 82.8% in the horizontal and vertical components. This abnormality lasts to the day after DOY340, which is consistent with the product accuracy evaluations in Section 2.2. The average convergence time is approximately 17.4 and 16.2 min for BDS-only and BDS + GPS during the whole test period, respectively, as shown in Fig. 9 and Table 5, with 81.8% and 73.6% shorter than 20 and 30 min. It is worthy to note that the accuracy threshold for BDS-only and BDS + GPS PPP convergence is different, namely, 0.3 and 0.6 m for BDS-only while 0.2 and 0.4 m for BDS + GPS PPP in the horizontal and vertical component, respectively (see Fig. 10).

4.2. Kinematic PPP for vehicle experiment

In addition to the simulated kinematic PPP at the six permanent stations, we also conducted two vehicle tests. The first test (test #1) was carried out on Jul 24, 2020, with a duration of about 2.5 h, and the second test (test #2) was on Dec 22, 2020, with a duration of about 1.5 h. The GNSS antenna was mounted on the roof of the vehicle, and the environment is overall open and unobstructed as shown

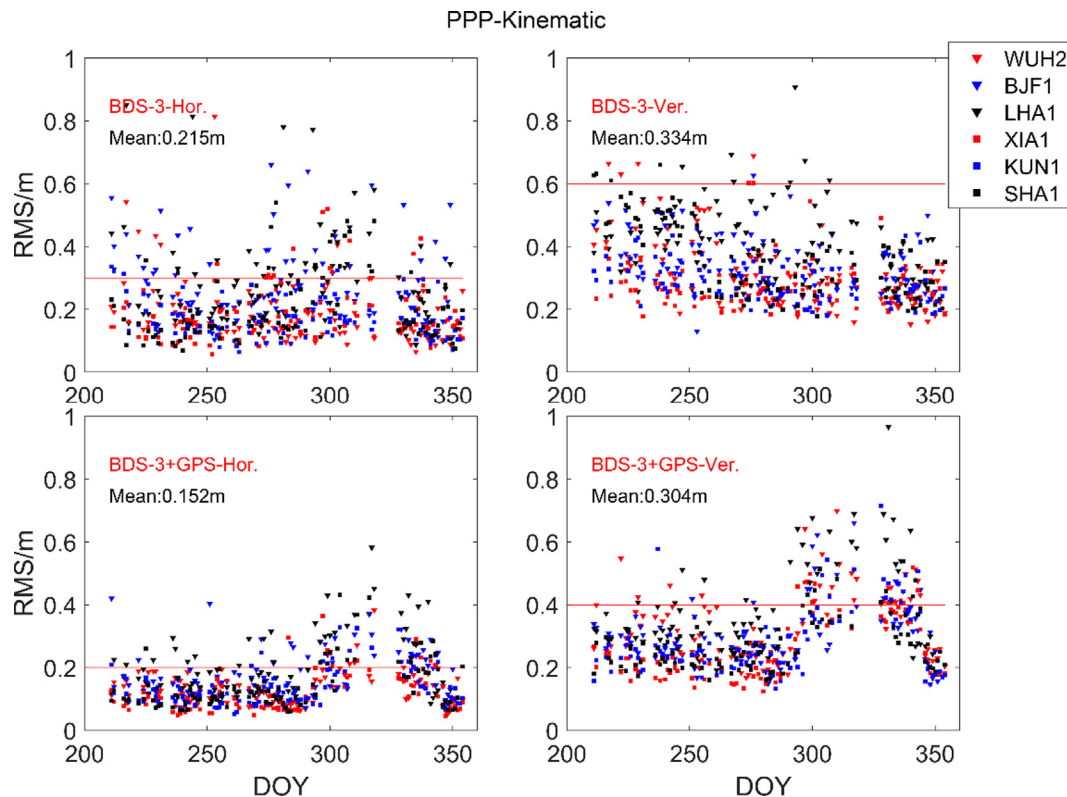


Fig. 8. Kinematic PPP position errors (95%) for BDS-only and BDS + GPS at six permanent stations. The horizontal red lines denote the threshold value for convergence. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 5

Kinematic PPP position error and convergence time at permanent stations.

Stations	BDS-only (95%)		GPS + BDS (95%)		Convergence time (min)	
	Hor. (m)	Ver. (m)	Hor. (m)	Ver. (m)	BDS	BDS + GPS
BJF1	0.274	0.342	0.176	0.302	12.2	13.2
LHA1	0.277	0.419	0.203	0.378	16.9	13.9
KUN1	0.170	0.290	0.139	0.286	14.5	15.9
WHU2	0.185	0.315	0.135	0.327	21.8	18.0
SHA1	0.198	0.356	0.142	0.272	19.3	18.3
XIA1	0.188	0.280	0.117	0.255	19.9	18.2
Mean	0.215	0.334	0.152	0.303	17.4	16.3

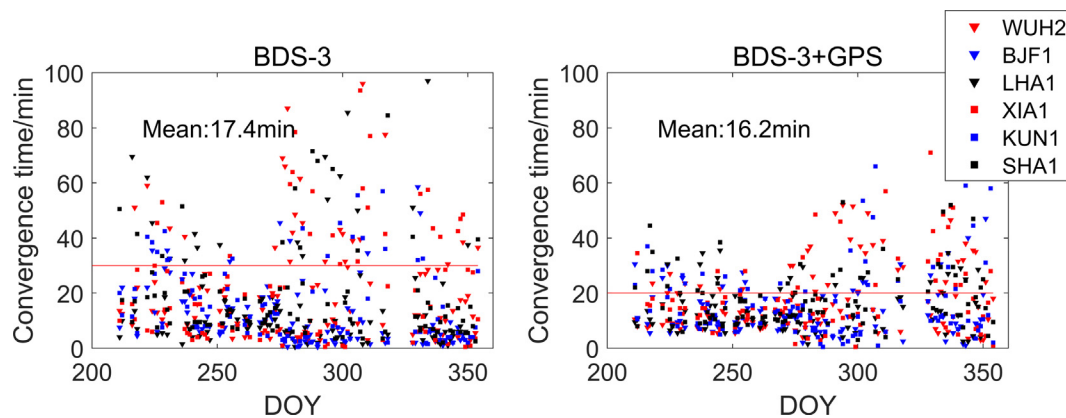


Fig. 9. Kinematic PPP convergence time for BDS-only and BDS + GPS at six permanent stations. The horizontal red lines denote 30 min and 20 min for BDS-only and BDS + GPS, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)



Fig. 10. Vehicle test scenario.

in Fig. 10. Observations were saved in the receiver and then were exported to the PC and processed in the kinematic PPP mode. Taking the NRTK results as a reference, the PPP position error and convergence time are then analyzed.

As can be seen from the Fig. 11 for test #1, the number of the visible satellites experiences a significant drop at around 02:35 UTC, which is due to the passing of a short tunnel, causing the re-convergence shown in both the BDS-only and GPS + BDS PPP position error time series. The results before the convergence (the criteria is the same as defined in Section 4.1) are removed, and the positioning

error (95%) is 27.3 and 49.9 cm in the horizontal and vertical respectively for BDS-only PPP, and is 17.8 and 36.3 cm for BDS + GPS PPP. The convergence time is approximately 10.4 and 14.2 min for BDS-only and BDS + GPS (see Fig. 12).

For vehicle test #2 as shown in Fig. 12, the positioning error (95%) excluding the pre-convergence period is 19.6 and 47.6 cm in the horizontal and vertical respectively for BDS-only PPP, and is 19.4 and 37.9 cm for BDS + GPS PPP. The convergence time is approximately 27.1 and 19.9 min.

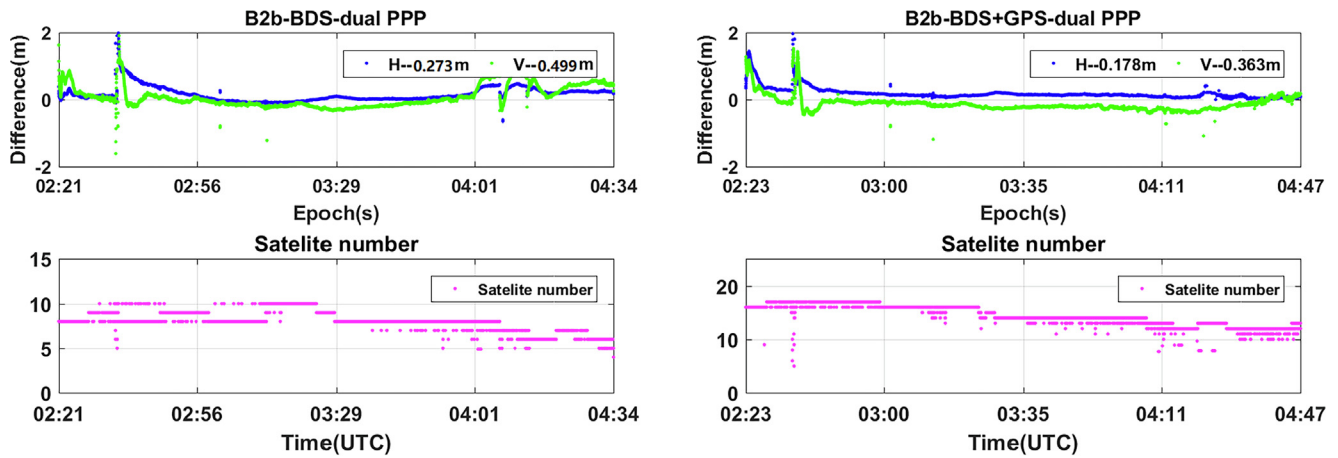


Fig. 11. BDS-only (left) and BDS + GPS (right) kinematic PPP results for test #1.

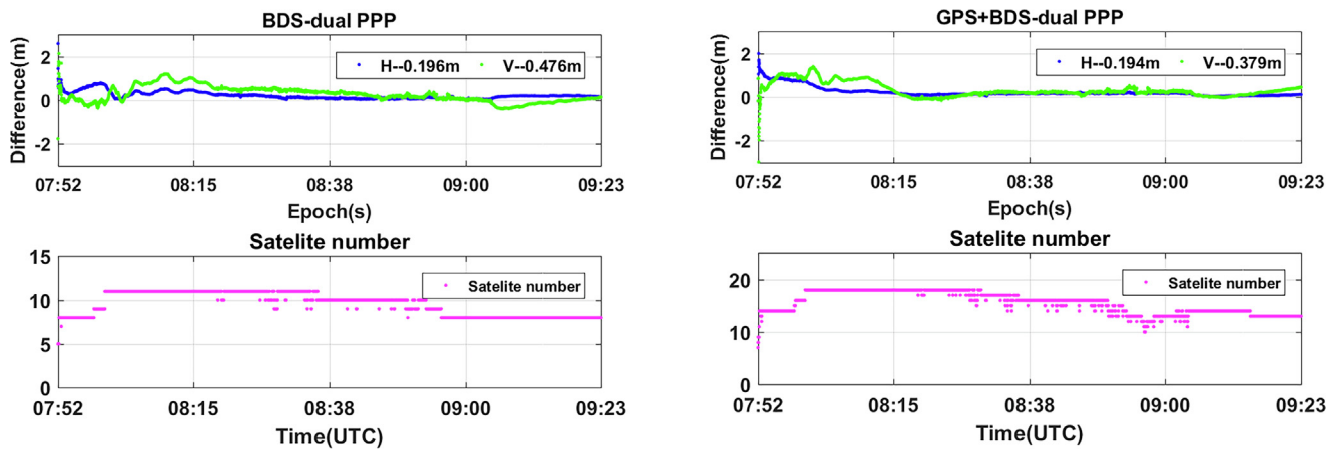


Fig. 12. BDS-only (left) and BDS + GPS (right) kinematic PPP results for test #2.

5. Conclusions

The China BDS-3 system has started to provide the free and open PPP service to users in China and surrounding areas since July 2020, by transmitting corrections for both BDS-3 and GPS satellites. This work carried out the initial assessment of the BDS-3 PPP service from the perspective of both the disseminated products and the PPP performance. The average availability of orbit corrections, clock corrections and DCB for BDS-3 satellites is about 36.31, 35.21 and 36.30%, respectively, due to the fact that the products are based on a regional ground network. Taking the GFZ GBM products as reference, the B2b orbit error RMS is about 10.2, 31.8 and 27.7 cm in average during the near five-month evaluation period in the radial, along and cross component respectively for BDS-3. The GPS orbit accuracy experiences a sudden decrease in orbit along and cross component after Oct. 22, 2020, resulting in relatively larger orbit errors compared to BDS-3, with average RMS of 7.2, 104.7 and 43.8 cm in the R, A and C components respectively. The clock error RMS and STD are approximately 2.936 and 0.243 ns for BDS-3, and about

5.089 and 0.285 ns for GPS. The SISRE, which combines the impact of both orbit and clock errors, is about 82.3 cm (RMS) and 3.9 cm (STD) for BDS-3, and about 135.8 cm (RMS) and 7.5 cm (STD) for GPS.

The static PPP based on B2b products was conducted at six permanent GNSS stations in China. The average position error RMS is about 2.1 and 2.6 cm in the horizontal and vertical component for BDS-only PPP, compared to 1.8 and 2.1 cm for the combined GPS and BDS-3 PPP. The measurements were also processed in simulated kinematic mode at these six stations, resulting in about 21.5 and 33.4 cm (95%) in the horizontal and vertical component for BDS-only PPP, with an average convergence time of 17.4 min. For combining BDS-3 and GPS, the average PPP position error is about 15.2 and 30.3 cm, and the convergence time is 16.3 min. Two vehicle-based PPP tests show average position error (95%) of about 23.5 (18.6) and 48.8 (37.1) cm in horizontal and vertical respectively, with a convergence time of 18.8 (17.1) min for BDS-only (BDS + GPS) PPP.

The evaluation results are overall in accordance with the claim of centimeter-level in static mode and decimeter-level

in the kinematic mode for BDS-3 PPP service. The BDS-3 PPP service is under extension with larger bandwidth to pursue better performance. Together with free PPP signals from other GNSS systems, such as the Galileo and QZSS, more robust and accurate PPP can be expected in the near future, which would greatly prosper the related applications.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The GBM products were accessed from <https://dataservices.gfz-potsdam.de>. The IGS MGEX observations can be downloaded from <https://gdc.cddis.eosdis.nasa.gov>. The iGMAS observations can be applied from <http://en.igmas.org/>. The vehicle test data can be freely obtained upon request.

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